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Impact of Allergic Rhinitis-Induced Sleep Disruption on Cognition and Physical Performance in Athletes: A Narrative Review

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Abstract

Background: Allergic rhinitis (AR) is a pervasive condition among competitive athletes, especially in endurance and aquatic disciplines. IgE-mediated Th2-type inflammation causes mucosal edema and increased upper airway resistance, forcing nocturnal mouth-breathing mechanics. This instability triggers frequent micro-arousals and disrupts sleep architecture, leading to documented reductions in restorative slow-wave and rapid eye movement sleep.

Aim: To synthesize evidence on how AR-mediated sleep fragmentation degrades neurocognitive function, motor control, and physical output in athletes, while evaluating the safety and efficacy of contemporary pharmacotherapy.

Material and methods: Narrative review integrating clinical, immunological, somnological, and sports science research from primary databases, including PubMed and Scopus.

Results: Systemic pro-inflammatory cytokines and nocturnal airway obstruction disrupt sleep-wake regulation centers. Resulting fragmentation impairs executive function, spatial working memory, and reaction times. Physiological decrements manifest as reduced aerobic efficiency, decreased explosive power, and diminished agility, often accompanied by an elevated rate of perceived exertion. While sedating first-generation antihistamines further impair performance and sleep architecture, intranasal corticosteroids and second-generation antihistamines provide effective symptom relief without cognitive impairment or World Anti-Doping Agency compliance issues.

Conclusions: AR-induced sleep disruption is a significant, underdiagnosed performance-limiting factor. Proactive allergy screening, longitudinal sleep tracking, and the optimization of non-sedating therapeutic regimens are essential for safeguarding athlete health and maintaining the competitive edge.

Key words: allergic rhinitis, sleep disruption, athletic performance, executive function, sports medicine

Introduction

Allergic rhinitis (AR) is a pervasive immunoglobulin E (IgE)-mediated chronic inflammatory condition, affecting approximately 10% to 40% of the global population (Tabata et al., 2025; Feng et al., 2026; Yan et al., 2025; Liu et al., 2020). Within competitive athletic cohorts, the prevalence is

frequently reported to be significantly higher than in the general public, ranging from 21% to 56.5% across diverse sporting disciplines (Matyja et al., 2025; Surda et al., 2017). This incidence peaks in specialized training environments; for instance, high-level swimmers and winter sports athletes demonstrate rhinitis prevalence rates as high as 40% to 74% (Tabata et al., 2025; Matyja et al., 2025; Bougault et al., 2010). Within elite Olympic populations, systematic clinical evaluations have identified asthma in 13.3% of athletes, of whom over 55% suffer from concomitant AR, while isolated allergic rhinitis affects approximately 14.8% of the total athletic population (Kurowski et al., 2014; Kurowski et al., 2016). The burden of this condition is further exacerbated by the high ventilatory demands required during intense exercise, which facilitate deeper penetration of aeroallergens and pollutants into the respiratory tract, leading to mucosal damage and epithelial barrier defects (Boulet and Turmel, 2017; Kurowski et al., 2014; Zaccarin et al., 2022). Consequently, AR presents a substantial socioeconomic and personal burden, characterized by reduced health-related quality of life, impaired social functioning, and significant productivity losses due to both presenteeism and absenteeism (Tabata et al., 2025; Rollema et al., 2022; Hossenbaccus et al., 2020; Feng et al., 2026).

Historically, allergic rhinitis has often been misclassified within sports medicine as a "minor nuisance" or a simple "stuffy nose," leading to widespread underdiagnosis and suboptimal self-management (Irmak et al., 2026; Bridgeman, 2017; Papapostolou et al., 2021). Data suggest that only 22% of symptomatic individuals receive a formal physician diagnosis, indicating that the true impact of AR on athletic readiness and recovery has been chronically underestimated (Bridgeman, 2017; Nathan et al., 2008). Traditional perspectives in sports medicine have long prioritized diet and exercise as the primary cornerstones of health, often overlooking the critical restorative role of sleep as an essential third pillar (Vitale et al., 2019; Sirohi et al., 2022). This historical trivialization has frequently led to the over-utilization of older, sedating first-generation antihistamines, which further compromise neurocognitive function, reaction time, and athlete safety (Fein et al., 2019; Papapostolou et al., 2021; Hossenbaccus et al., 2020). Modern evidence-based frameworks now emphasize that uncontrolled AR functions as a primary performance-limiting condition that impairs breathing efficiency and the achievement of peak physical output, which is predicated on optimal respiratory health (Matyja et al., 2025; Papapostolou et al., 2021; Choi et al., 2025).

A critical paradigm shift in sports medicine recognizes a complex "neural-immune bridge" connecting mucosal inflammation to neurocognitive and physical decline (Feng et al., 2026; Adeniji et al., 2026). Nasal congestion, the most bothersome symptom of AR, increases upper airway resistance and functions as a primary risk factor for Sleep-Disordered Breathing (SDB) and frequent nocturnal micro-arousals (Irawati et al., 2023; Feng et al., 2026; Liu et al., 2020). Pro-inflammatory

cytokines released during the allergic response, such as IL-1 β , IL-6, and TNF- α , circulate systemically and enter the central nervous system, where they act on brain regions involved in emotion and sleep regulation, including the hippocampus and amygdala (Feng et al., 2026; Adeniji et al., 2026). This inflammatory cascade disrupts the hypothalamic-pituitary-adrenal (HPA) axis and alters neurotransmitter balance, resulting in neurocognitive fatigue, decreased cognitive flexibility, and impaired memory consolidation (Feng et al., 2026; Sun et al., 2025; Adeniji et al., 2026). For the elite athlete, this disruption to sleep architecture manifests as prolonged reaction times, diminished neuromuscular coordination, and reduced time to exhaustion, directly compromising tactical execution and physical performance in both endurance and combat contexts (Wannell et al., 2025; Amri et al., 2026; Vitale et al., 2019; Song et al., 2025).

Research Objective

The research objective of this review is to provide a comprehensive synthesis of clinical and neurobiological evidence to delineate the multifaceted pathways through which AR-induced nocturnal airway resistance and systemic inflammation compromise athletic performance. By integrating findings across immunology, somnology, and sports science, the review aims to establish a framework for recognizing AR as a primary performance-limiting disorder. The objective further extends to evaluating the efficacy of current therapeutic interventions, specifically intranasal corticosteroids and newer-generation antihistamines, in restoring sleep quality and cognitive sharpness while adhering to international anti-doping regulations (Tabata et al., 2025; Irmak et al., 2026; Boulet and Turmel, 2017; Rollema et al., 2022).

Research Problems

Dedicated research problems addressed in this review involve identifying the specific pathophysiological pathways linking AR-mediated Th2-type inflammation to the fragmentation of restorative slow-wave and REM sleep. How does the systemic elevation of cytokines like IL-6 and TNF- α interfere with the synthesis of melatonin and the stability of the thalamocortical circuit in athletes? To what extent does this neurocognitive degradation translate into objective declines in explosive power, agility, and motor control during high-intensity competition? Furthermore, what are the current challenges in optimizing safe pharmacotherapy that manages nocturnal symptoms without inducing secondary daytime drowsiness or violating WADA 2025 regulatory updates (Feng et al., 2026; Sun et al., 2025; Amri et al., 2026; Hostrup et al., 2023; Vitale et al., 2019)?

Research Hypotheses

The research hypotheses for this review posit that chronic AR-induced nasal obstruction causes significant disruption to sleep architecture, which serves as the primary mediator of cognitive fatigue and reduced attentional control in elite athletes. It is hypothesized that athletes with higher symptom severity possess a smaller "cognitive buffer," making them more vulnerable to performance failures under the acute stress of competition. We further hypothesize that the inflammatory sleep-wake cycle disruption creates a bidirectional vicious cycle, where inadequate sleep exacerbates systemic mucosal inflammation, thereby deepening the performance deficit. Finally, we hypothesize that the implementation of standardized intranasal corticosteroid therapy will result in superior improvements in reaction time and physical output compared to antihistamine monotherapy by more effectively alleviating nocturnal airway resistance (Song et al., 2025; Feng et al., 2026; Lu et al., 2025; Irmak et al., 2026; Tabata et al., 2025).

Research materials and methods

Literature for this review was identified through systematic searches across primary biomedical and sports science databases, including PubMed/MEDLINE and Scopus. The search protocols employed Boolean operators to integrate MeSH terms and keywords encompassing allergic rhinitis, hay fever, and IgE-mediated inflammation in conjunction with sleep-specific variables such as sleep architecture, slow-wave sleep, rapid eye movement cycles, and nocturnal micro-arousals. To capture the full spectrum of the performance deficit, search strings also included terms related to executive function, cognitive flexibility, reaction time, motor control, and various indices of athletic output in both endurance and high-intensity sporting contexts.

Explicit selection criteria were applied to ensure the clinical relevance and academic rigor of the synthesized evidence. Inclusion was restricted to peer-reviewed original research, randomized controlled trials, systematic reviews, meta-analyses, and official position statements. Eligible studies were required to focus on athletic or physically active populations diagnosed with IgE-mediated allergic rhinitis through validated clinical evaluations, skin prick testing, or serum-specific IgE assessments. Furthermore, selected papers had to report on either objective sleep parameters via polysomnography and actigraphy or subjective metrics through validated instruments, alongside standardized testing of neurocognitive function or physical performance metrics. In alignment with modern sports medicine requirements, specific consideration was given to studies addressing anti-doping compliance and pharmacological safety. Exclusion criteria strictly eliminated non-peer-reviewed literature, conference abstracts, and studies focusing solely on non-allergic anatomical nasal defects, such as structural polyps or septal deviation, without a chronic inflammatory etiology.

Animal-only studies without clear human translational significance and papers lacking adequate methodological transparency or rigorous statistical controls were also omitted from the final synthesis.

Data extraction and analytical synthesis followed a systematic thematic framework to delineate the multifaceted pathways linking mucosal inflammation to performance decline. Findings were categorized across several critical domains: the pathophysiological cascades involving Th2-type cytokines and their systemic influence on the central nervous system; the neuroimmunological disruption of the hypothalamic-pituitary-adrenal axis and circadian rhythms; and the specific mechanisms of sleep architecture fragmentation and non-restorative rest. Qualitative synthesis further organized data regarding the degradation of executive functions, including attentional control and spatial working memory, as well as objective declines in explosive power, agility, and sport-specific motor skills. Finally, evidence concerning pharmacological management was synthesized to contrast the efficacy of various therapeutic interventions with their safety profiles under current World Anti-Doping Agency regulatory updates.

The identified evidence was critically appraised for thematic consistency, clinical relevance to elite and competitive sport, and overall methodological validity to construct a rigorous narrative synthesis. Discrepancies in findings across different sporting training environments, such as swimming and winter sports, were evaluated based on exposure-specific inflammatory loads and environmental triggers. By assessing the quality of the literature through established frameworks for evaluating risk of bias and the certainty of evidence, this review integrates diverse findings into a cohesive narrative. This process ensured that the resulting synthesis provides a robust, evidence-based understanding of the systemic interactions between allergic rhinitis, restorative sleep, and the preservation of peak athletic readiness.

AI was utilized for two specific purposes in this research. Text analysis of clinical reasoning narratives to identify linguistic patterns associated with specific logical fallacies. Assistance in refining the academic English language of the manuscript, ensuring clarity, consistency, and adherence to scientific writing standards. AI were used for additional linguistic refinement of the research manuscript, ensuring proper English grammar, style, and clarity in the presentation of results. It is important to emphasize that all AI tools were used strictly as assistive instruments under human supervision. The final interpretation of results, classification of errors, and conclusions were determined by human experts in clinical medicine and formal logic. The AI tools served primarily to

enhance efficiency in data processing, pattern recognition, and linguistic refinement, rather than replacing human judgment in the analytical process.

Research results

Pathophysiological Cascades and Nocturnal Airway Resistance

Allergic rhinitis (AR) is fundamentally defined by a chronic immunoglobulin E (IgE)-mediated Th2-type inflammatory response of the nasal mucosa (Tabata et al., 2025; Feng et al., 2026). Upon aeroallergen exposure, a complex molecular cascade is initiated, characterized by the local and systemic release of inflammatory mediators including histamine, cysteinyl leukotrienes, and pro-inflammatory cytokines such as IL-4, IL-5, and IL-13 (Agnihotri and McGrath, 2019; Feng et al., 2026; Hossenbaccus et al., 2020). These mediators act synergistically to induce significant nasal mucosal edema, vascular engorgement, and goblet cell hyperplasia, which collectively increase upper airway resistance (Irawati et al., 2023; Feng et al., 2026; Kwiliński et al., 2024). This physiological burden is uniquely amplified in competitive athletes; the large ventilatory volumes required during high-intensity training facilitate deeper penetration of allergens and environmental pollutants into the lower respiratory tract, leading to significant airway dehydration and epithelial barrier defects (Boulet and Turmel, 2017; Kurowski et al., 2014).

The mechanical obstruction of the nasal passages necessitates a compensatory shift from physiological nasal breathing to forced nocturnal mouth-breathing mechanics (Feng et al., 2026; Oliveira et al., 2025). Mouth breathing significantly destabilizes the upper airway during sleep by promoting the posterior displacement of the tongue root and decreasing the longitudinal tension of the pharyngeal dilator muscles, thereby increasing the risk of airway collapse and Sleep-Disordered Breathing (SDB) (Feng et al., 2026; Oliveira et al., 2025; Irawati et al., 2023). Furthermore, the circadian decline in serum cortisol levels during the night and early morning, which normally serves to suppress inflammatory processes, exacerbates nasal resistance and congestion, leading to a zenith of symptoms that peak just as athletes begin early morning training sessions (Irawati et al., 2023; Papapostolou et al., 2021).

Beyond mechanical factors, a "neural-immune bridge" allows systemically circulating inflammatory markers, specifically IL-1 β , IL-6, and TNF-alpha, to cross the blood-brain barrier via active transport or the vagus nerve afferent pathway (Feng et al., 2026; Adeniji et al., 2026). Once within the central nervous system, these cytokines act directly on neuroanatomical regions responsible for sleep-wake regulation, such as the suprachiasmatic nucleus (SCN), the hypothalamus, and the hippocampus (Feng et al., 2026; Adeniji et al., 2026). In aquatic athletes, these cascades are further intensified by exposure to water disinfection byproducts like trichloramine, which induce mucosal damage and have

been associated with a significant association between training hours per session and nasal congestion in winter months (OR = 3.10; p = 0.039) (Matyja et al., 2025; Zaccarin et al., 2022).

Sleep Architecture Disruption and Nocturnal Micro-Arousals

The objective evaluation of sleep parameters in AR patients through polysomnography (PSG) reveals profound alterations in sleep architecture, primarily manifesting as significant reductions in restorative Slow-Wave Sleep (SWS or N3) and Rapid Eye Movement (REM) sleep (Irawati et al., 2023; Feng et al., 2026). Chronic nasal obstruction and the resulting increased respiratory effort trigger frequent nocturnal micro-arousals and Respiratory Effort-Related Arousals (RERAs), which are documented to be significantly higher in allergic cohorts compared to matched non-allergic controls (Irawati et al., 2023; Liu et al., 2020). Quantitative analysis indicates that individuals with moderate-to-severe AR experience a reduction of approximately 38.5 ± 6.2 minutes in slow-wave sleep and a corresponding prolongation of REM sleep latency by over 20 minutes (Feng et al., 2026; Thompson et al. in Feng et al., 2026).

These architectural disruptions severely compromise essential muscular and anabolic recovery pathways. SWS is recognized as a critical phase for anabolic metabolism and macromolecule RNA synthesis, during which approximately 70% of daily growth hormone (GH) secretion occurs (Amri et al., 2026; Irawati et al., 2023). Fragmented sleep prevents the athlete from reaching and maintaining these deep stages, thereby impeding muscle protein synthesis, tissue repair, and immune system reinforcement (Amri et al., 2026; Irawati et al., 2023; Vitale et al., 2019). Furthermore, the allergic inflammatory state creates a bidirectional vicious cycle: intermittent hypoxia from SDB and sleep fragmentation act as physiological stressors that activate the hypothalamic-pituitary-adrenal (HPA) axis, further increasing systemic IL-6 and TNF-alpha levels, which in turn inhibit the synthesis and secretion of melatonin by the pineal gland (Feng et al., 2026; Sun et al., 2025; Adeniji et al., 2026). Clinical data show that the nocturnal peak serum melatonin level in AR patients is 27.3% lower than in healthy controls, directly correlating with poorer sleep quality (Feng et al., 2026).

The impact on REM sleep is particularly significant for athletes involved in skill-based and tactical sports, as REM sleep is vital for memory consolidation and the refinement of procedural learning (Oliveira et al., 2025; Adeniji et al., 2026). Allergic individuals demonstrate a decrease in total percent time in REM sleep to as low as 10.9% compared to normal physiological expectations (Irawati et al., 2023). This non-restorative sleep manifests as excessive daytime sleepiness (EDS) and morning headache, which increase the risk of overtraining and accidental injuries (Liu et al., 2020; Vitale et al., 2019; Amri et al., 2026).

Neurocognitive Impairment and Executive Function Degradation

The convergence of systemic neuroinflammation and non-restorative sleep induces significant neurocognitive deficits that directly compromise athletic readiness. Objective testing using the Cambridge Neuropsychological Test Automated Battery (CANTAB) has demonstrated impaired spatial working memory and significantly higher error rates during peak pollen seasons in children and adolescents with AR (Papapostolou et al., 2021). Allergic inflammation is associated with a "feedback blunting" effect, where the brain's ability to flexibly adapt to changing conditions and update strategies based on performance cues is severely diminished (Sun et al., 2025; Whitney et al., 2017). This executive function degradation is linked to reduced cerebrovascular reactivity (CVR) in the temporal-occipital cortex and a lower "cognitive buffer," making athletes more vulnerable to decision-making failures under high-intensity competition stress (Sun et al., 2025; Song et al., 2025; Adeniji et al., 2026).

Psychomotor performance is markedly deteriorated, with athletes exhibiting significant prolongations in both simple reaction time (SRT) and multiple-choice reaction time (MCRT) following nights of restricted sleep (Amri et al., 2026; Wannell et al., 2025). In international-level karate athletes, sleep restriction to 4 hours per night resulted in a significant increase in Stroop reaction times ($F = 13.0$, $P < 0.01$), with further deterioration observed after physical exertion, indicating that cognitive fatigue is compounded by exercise-induced strain (Amri et al., 2026). Visual tracking, attentional control, and information processing speed are also compromised, as evidenced by reduced delta frequency power in the central brain regions and altered P3 event-related potential (ERP) amplitudes (Sun et al., 2025; Song et al., 2025; Adeniji et al., 2026). These neural biomarkers reflect a diminished capability to manage stress and predict competition outcomes (Song et al., 2025).

The neurocognitive impact extends to behavioral symptoms of irritability, shyness, and social withdrawal, which are particularly prominent in adolescent athletes (Feng et al., 2026; Irmak et al., 2026). Sleep loss inhibits prefrontal cortical function, leading to decreased emotional regulation and an increased risk of anxiety and depression, which 42% of the effect between AR and mood is mediated by poor sleep quality (Feng et al., 2026). For the elite athlete, these deficits translate to longer reaction times for simple movement and a reduced "cognitive reserve" that impairs tactical execution in sports where split-second decisions determine victory (Papapostolou et al., 2021; Amri et al., 2026; Song et al., 2025).

Direct and Indirect Athletic Performance Decrements

The impact of AR-induced sleep disruption on physical output encompasses a broad range of endurance, neuromuscular, and motor control metrics. Allergic athletes frequently experience seasonal performance dips during peak allergen exposure, where increased symptom severity

correlates with objective declines in training load tolerance and higher rates of upper respiratory tract symptoms (Song et al., 2025; Zaccarin et al., 2022). Direct performance decrements have been quantified in elite athletes following partial sleep restriction, including significant declines in karate-specific aerobic test (KSAT) endurance and a shortened time to exhaustion in standardized treadmill protocols (Amri et al., 2026; Vitale et al., 2019). High-intensity motor control is similarly compromised, with documented reductions in squat jump (SJ) height ($F = 15.8$, $P < 0.001$) and significantly poorer performance in the modified agility T-test (MAT) ($F = 28.5$, $P < 0.001$) (Amri et al., 2026).

These physiological declines are often accompanied by an elevated rate of perceived exertion (RPE), suggesting that athletes must exert greater psychological and physical effort to maintain submaximal aerobic efficiency when sleep is fragmented (Vitale et al., 2019; Wannell et al., 2025; Sirohi et al., 2022). In endurance cyclists and runners, one night of restricted sleep (24 h wakefulness) has been shown to reduce total running distance by significant margins and attenuate blood pressure responses to cycling exercise (Vitale et al., 2019; Wannell et al., 2025). In swimming cohorts, training in chlorinated environments induces epithelial damage that exacerbates winter-season symptoms, with significant increases in coughing ($OR = 3.48$; $p = 0.015$) and nasal congestion that directly interfere with breathing efficiency during training (Matyja et al., 2025; Zaccarin et al., 2022).

Conversely, structured aerobic exercise has been explored as a potential immunomodulatory adjunct for AR management. Eight weeks of moderate-intensity training can improve VO_2 max, increase peak nasal inspiratory flow (PNIF), and reduce eosinophilic airway inflammation as measured by fractional exhaled nitric oxide (FeNO) (Chanta et al., 2026). However, high-intensity exercise may also induce a transient immunosuppression or increase allergen exposure, which can exacerbate symptoms if not managed correctly (Yan et al., 2025). The overall training state is compromised by a reduction in total sleep duration, which in adolescents with AR was significantly lower ($p = 0.001$) compared to healthy peers, contributing to persistent fatigue and reduced training capacity (Choi et al., 2025; Vitale et al., 2019).

Pharmacological Safety, Anti-Doping Regulations, and Cognitive Profiles

Pharmacotherapeutic management in the athletic population is governed by the critical need to balance symptom control with cognitive safety and compliance with the 2025 World Anti-Doping Agency (WADA) Prohibited List (Hostrup et al., 2023; Michalik, 2025; Boulet and Turmel, 2017). Older, first-generation antihistamines (e.g., diphenhydramine, chlorpheniramine) are strictly discouraged due to their potent central nervous system (CNS) sedative effects and their ability to cross the blood-brain barrier easily (Fein et al., 2019; Bridgeman, 2017; Hossenbaccus et al., 2020). These agents disrupt sleep architecture by reducing REM sleep and producing next-day "hangover"

symptoms such as impaired vigilance, divided attention, and compromised sensory-motor performance (Fein et al., 2019; Bridgeman, 2017). In contrast, second-generation (non-sedating) antihistamines and intranasal corticosteroids (INCS) demonstrate a favorable cognitive safety profile and are identified as first-line treatments for restoring sleep quality and improving disease-specific quality of life (Tabata et al., 2025; Irmak et al., 2026; Rollema et al., 2022).

Intranasal corticosteroids (e.g., fluticasone, mometasone, ciclesonide) are identified as the most effective monotherapy for moderate-to-severe AR, specifically for alleviating nocturnal airway resistance and reducing the severity of SDB (Tabata et al., 2025; Rollema et al., 2022; Hossenbaccus et al., 2020). Meta-analysis data from 18 RCTs involving over 6,000 participants show that INCS significantly improve sleep quality across standardized metrics, with clinically meaningful improvements often exceeding the minimal clinically important difference (MCID) (Tabata et al., 2025). Under current WADA 2025 regulations, all intranasal corticosteroids are permitted in and out of competition without the need for a Therapeutic Use Exemption (TUE) (Boulet and Turmel, 2017; World Anti-Doping Agency, 2024; Michalik, 2025).

Practitioners must remain vigilant regarding comorbid asthma treatments; while inhaled corticosteroids (ICS) and most second-generation antihistamines are permitted, specific beta-2 agonists (e.g., salbutamol, formoterol) are subject to strict dose-limited conditions, and systemic (oral) corticosteroids remain prohibited in-competition (Boulet and Turmel, 2017; Hostrup et al., 2023; World Anti-Doping Agency, 2024). Emerging biological therapies such as omalizumab and dupilumab, which target IgE and IL-4/IL-13 respectively, are permitted and represent a significant advance for athletes with refractory or moderate-to-severe persistent AR who remain symptomatic on standard regimens (Magierska et al., 2024; Feng et al., 2026; Michalik, 2025). Optimal management should prioritize INCS monotherapy to enhance alertness and training productivity while ensuring total anti-doping compliance (Irmak et al., 2026; Fein et al., 2019; Boulet and Turmel, 2017).

Table 1. Impact of Allergic Rhinitis on Key Sleep Parameters and Cognitive Metrics

Metric Assessed	AR / Symptomatic Impact	Clinical / Performance Interpretation
Slow-Wave Sleep (SWS)	38.5 minute reduction	Impaired anabolic muscle recovery
REM Sleep Percentage	10.3% – 10.9% total time	Deteriorated procedural memory consolidation
Sleep Efficiency	3.95% overall decrease	Lowered restorative sleep quality
Spatial Memory Errors	92% increase in errors	Significant tactical decision-making failure

Simple Reaction Time	29 millisecond delay	Slower motor response to stimuli
Choice Reaction Time	70 millisecond delay	Impaired cognitive processing speed
Peak Melatonin Levels	27.3% lower peak concentration	Disrupted circadian sleep-wake cycle

Table 2. Physiological and Physical Performance Changes in Symptomatic Athletes

Performance Indicator	Measured Quantitative Change	Athletic Consequence
Aerobic Endurance (KSAT)	11.9% reduction in duration	Shortened time to exhaustion
Explosive Power (Squat Jump)	3.5% reduction in height	Decreased vertical force production
Agility (MAT Protocol)	5.4% performance decline	Slower multi-directional motor control
Training Productivity	25% reduction on peak days	Lowered session training efficiency
Nasal Congestion Risk	3.10 odds ratio (increase)	Impaired respiratory training mechanics
Airway Inflammation (FeNO)	33.3% higher baseline levels	Elevated systemic inflammatory load
Running Distance Covered	3.0% reduction in total volume	Diminished submaximal aerobic output

Discussion

The biological interaction between allergic rhinitis and athletic readiness is underpinned by a complex cascade of mechanical and immunological factors that validate the link between mucosal distress and performance decline. Chronic mucosal inflammation, driven by Th2-type cytokines, results in persistent nasal congestion that significantly elevates upper airway resistance during recumbency. This physiological shift necessitates a transition to forced mouth breathing, which destabilizes the upper airway and acts as a primary catalyst for sleep-disordered breathing and frequent nocturnal micro-arousals (Feng et al., 2026; Irawati et al., 2023). The resultant fragmentation of sleep architecture, specifically the documented reductions in slow-wave sleep and rapid eye movement phases, impedes essential anabolic recovery, hormonal regulation, and neural processing (Adeniji et al., 2026; Oliveira et al., 2025). Consequently, the systemic elevation of pro-inflammatory markers acting on the central nervous system confirms a functional descent into neurocognitive fatigue. This deterioration is manifested through prolonged reaction times, impaired executive function, and

diminished spatial awareness, which collectively compromise physical output and tactical decision-making in elite performance environments (Amri et al., 2026; Sun et al., 2025; Wannell et al., 2025). Translating these pathophysiological insights into clinical practice requires a multidisciplinary framework focused on early identification and targeted intervention to preserve an athlete's "cognitive buffer." Sports medicine practitioners should prioritize the use of validated screening tools, such as the Allergy Questionnaire for Athletes (AQUA), to identify symptomatic individuals who frequently remain undiagnosed or suboptimally managed (Bonini et al., 2009; Kurowski et al., 2014; Bridgeman, 2017). Management strategies during peak pollen seasons or high-risk training periods, such as winter swimming training, must integrate pharmacological safety with performance optimization. Intranasal corticosteroids and non-sedating second-generation antihistamines serve as the cornerstone of therapy, offering significant improvements in sleep quality and disease-specific quality of life without the central nervous system depression associated with older sedative medications (Tabata et al., 2025; Irmak et al., 2026; Rollema et al., 2022). Furthermore, the implementation of structured sleep hygiene protocols, including the strategic use of daytime naps and the concept of "banking sleep" prior to anticipated periods of restriction, can significantly mitigate the cumulative effects of training-induced fatigue and mucosal stress (Vitale et al., 2019; Sirohi et al., 2022).

Despite the consistency of findings across various cohorts, current literature is characterized by several methodological constraints that necessitate a cautious interpretation of the absolute magnitude of performance loss. Many studies focusing on elite athletic populations are limited by relatively small sample sizes and a high degree of homogeneity, which may restrict the generalizability of results across diverse sporting disciplines, genders, and age groups (Amri et al., 2026; Wannell et al., 2025; Choi et al., 2025). Furthermore, a significant proportion of the available evidence relies on subjective sleep surveys and retrospective self-reporting rather than the gold-standard polysomnographic evaluation, potentially overlooking transient nocturnal micro-arousals or subtle changes in sleep efficiency (Liu et al., 2020; Irawati et al., 2023). The predominance of cross-sectional designs further limits the ability of researchers to determine definitive causal relationships or to characterize the long-term trajectory of symptoms across full training cycles and fluctuating seasonal allergen loads (Matyja et al., 2025; Zaccarin et al., 2022).

Addressing these gaps represents a critical priority for future research within the intersecting fields of sports immunology and somnology. Prospective longitudinal studies utilizing ambulatory polysomnography and actigraphy during high-intensity training camps are essential to clarify the direct relationship between allergic symptom severity, sleep fragmentation, and metabolic recovery. Future investigations should also focus on real-time cognitive monitoring during peak allergen exposure to identify neural biomarkers of performance failure, such as altered P3 event-related potential amplitudes and reduced delta frequency power (Song et al., 2025; Sun et al., 2025).

Additionally, examining the long-term disease-modifying effects of allergen immunotherapy and the potential performance-restoring role of biological therapies could provide more durable solutions for athletes with refractory moderate-to-severe persistent rhinitis (Magierska et al., 2024; Feng et al., 2026). Integrating biomarker-guided management, such as the routine monitoring of fractional exhaled nitric oxide or systemic cytokine levels, will enable practitioners to provide more personalized and effective care for the allergic athlete (Chanta et al., 2026; Yan et al., 2025).

Conclusions

The evidence synthesized in this review confirms that allergic rhinitis is not a benign mucosal nuisance but rather a systemic inflammatory disorder with profound implications for an athlete's physiological and neurocognitive readiness. The primary pathophysiological driver of performance decline is the chronic increase in nocturnal upper airway resistance, which necessitates a transition to forced mouth-breathing mechanics. This mechanical instability, coupled with the systemic influence of pro-inflammatory cytokines acting on the central nervous system, results in a significant fragmentation of restorative sleep architecture. Specifically, the documented reductions in slow-wave and rapid eye movement sleep impede essential anabolic recovery and neural processing. The consequent descent into neurocognitive fatigue manifests as prolonged reaction times, impaired executive function, and diminished spatial awareness, which collectively compromise tactical decision-making and physical output in high-stakes training and competition environments.

To mitigate these risks, sports medicine practitioners must adopt a proactive management framework that prioritizes early diagnostic screening and the optimization of safe, evidence-compliant pharmacotherapy. Utilizing validated instruments such as the Allergy Questionnaire for Athletes allows for the identification of a large cohort of symptomatic individuals who currently remain underdiagnosed or suboptimally managed. Clinical intervention should focus on the standardized application of intranasal corticosteroids and non-sedating second-generation antihistamines, which effectively alleviate nocturnal airway resistance and restore sleep quality without inducing the central nervous system depression or sleep architecture disruption associated with older sedative agents. Furthermore, integrating seasonal pollen monitoring and structured sleep hygiene protocols into routine athletic care is essential to preserve the cognitive buffer required for elite performance. By recognizing the allergic rhinitis-sleep-performance axis as a critical component of medical oversight, practitioners can safeguard both the long-term health of the athlete and their sustained competitive edge.

Disclosure

Supplementary Materials

Not applicable.

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Conflicts of Interest

The authors deny any conflict of interest.

References

1. Adeniji, K. O., Igado, O. O., Olopade, F. E., & Femi-Akinlosotu, O. M. (2026). Sleep deprivation and memory: A neurobiological perspective. *IBRO Neuroscience Reports*, 20, 170–180. <https://doi.org/10.1016/j.ibneur.2026.01.010>
2. Amri, A., Faleh, J., Ben Aissa, M., Ceylan, H. İ., Ben Hassen, S., Hmaidi, J., Guelmami, N., Dergaa, I., Ștefănică, V., & Souissi, N. (2026). Effects of sleep restriction on cognitive and physical performance in elite karate athletes: A randomized crossover study. *Medicine*, 105(11), e48064. <https://doi.org/10.1097/MD.00000000000048064>
3. Boulet, L.-P., Turmel, J., & Irwin, R. S. (2017). Cough in the athlete: CHEST guideline and expert panel report. *CHEST*, 151(2), 441–454. <https://doi.org/10.1016/j.chest.2016.10.054>
4. Bridgeman, M. B. (2017). Overcoming barriers to intranasal corticosteroid use in patients with uncontrolled allergic rhinitis. *Integrated Pharmacy Research and Practice*, 6, 109–119. <https://doi.org/10.2147/IPRP.S119463>
5. Choi, H., Seo, E., & Woo, J. (2025). Association of allergic conditions with adolescent sleep duration: A national survey. *Children*, 12(10), 1356. <https://doi.org/10.3390/children12101356>
6. de Oliveira, J. B., et al. (2025). Sleep and chronic rhinosinusitis with nasal polyposis. *Sleep Science*, 18(4), 443. <https://doi.org/10.1055/s-0045-1811199>
7. Fein, M. N., Fischer, D. A., O'Keefe, A., & Sussman, G. (2019). CSACI position statement: Newer generation H1-antihistamines are safer than first-generation H1-antihistamines. *Allergy, Asthma & Clinical Immunology*, 15(1), 61. <https://doi.org/10.1186/s13223-019-0375-7>
8. Feng, X., Gu, Z., & Wang, Y. (2026). The relationship between allergic rhinitis and sleep disorders and mental health. *Frontiers in Allergy*, 7, 1806555. <https://doi.org/10.3389/falgy.2026.1806555>
9. Hossenbaccus, L., Linton, S., Garvey, S., & Ellis, A. K. (2020). Allergic rhinitis. *Allergy, Asthma & Clinical Immunology*, 16(1), 39. <https://doi.org/10.1186/s13223-020-00436-x>

10. Hoyte, F. C. L., & Nelson, H. S. (2018). Recent advances in allergic rhinitis [version 1]. *F1000Research*, 7, 1333. <https://doi.org/10.12688/f1000research.15367.1>
11. Irawati, N., Wardani, R. S., Poerbonegoro, N. L., & Rahmawati, I. (2023). Sleep disturbance in chronic rhinitis: Evaluation of nasal symptoms, sleep disorder questionnaires, and sleep architecture in allergic and non-allergic rhinitis. *Indian Journal of Otolaryngology and Head & Neck Surgery*. <https://doi.org/10.1007/s12070-023-03783-5>
12. Irmak, G., et al. (2026). Impact of allergic rhinitis on school performance in adolescents. *Journal of Asthma and Allergy*. <https://doi.org/10.2147/JAA.S606463>
13. Kerdkaew, K., Rattanathamma, P., Tongtako, W., Mickleborough, T., & Jirapongsatorn, B. (2026). Effects of moderate-intensity aerobic exercise on clinical symptoms and physiological outcomes in young adults with persistent allergic rhinitis: A randomized controlled trial. *International Journal of Environmental Research and Public Health*, 23(5), 611. <https://doi.org/10.3390/ijerph23050611>
14. Kurowski, M., Jurczyk, J., Krysztofiak, H., & Kowalski, M. L. (2014). Association of serum Clara cell protein CC16 with respiratory infections and immune response to respiratory pathogens in elite athletes. *Respiratory Research*, 15, 45. <https://doi.org/10.1186/1465-9921-15-45>
15. Kwiliński, K., Kucharska, M., Szyszka, M., Wawrzyńska, B., Przybysz, P., Paruch, A., & Wadowska, M. (2024). Allergic rhinitis – a review of current diagnostic and therapeutic methods. *Quality in Sport*, 19, 54182. <https://doi.org/10.12775/QS.2024.19.54182>
16. Liu, J., Zhang, X., Zhao, Y., & Wang, Y. (2020). The association between allergic rhinitis and sleep: A systematic review and meta-analysis of observational studies. *PLoS ONE*, 15(2), e0228533. <https://doi.org/10.1371/journal.pone.0228533>
17. Lu, Z., Li, S., et al. (2025). Joint association of sleep quality and physical activity with allergic rhinitis in preschool children. *Journal of Asthma and Allergy*, 18, 101. <https://doi.org/10.2147/JAA.S101>
18. Magierska, A., Szczepaniak, Z., Ptak, O., & Gomułka, K. (2024). Biological therapy, therapeutic approaches and future perspectives in the treatment of chronic rhinitis. *Quality in Sport*, 18, 52161. <https://doi.org/10.12775/QS.2024.18.52161>
19. Matyja, M., Kowalska, M., Mazur, A., Brodziak, J., Fojcik, K., Ciszewska, W., Kosztyła-Czech, Z., Wiśniewski, T., & Dworak, M. (2025). Rhinitis in swimmers: A clinical issue at the intersection of sports medicine and otorhinolaryngology. *Quality in Sport*, 47, 66927. <https://doi.org/10.12775/QS.2025.47.66927>
20. Michalik, M., Lorenc, T., Muras, M., Marcinkowski, K., Mikszta, N., Mikszta, J., & Marcinkowska, J. (2025). Key updates in the WADA 2025 Prohibited List: Implications for competitive sports. *Quality in Sport*, 42, 61192. <https://doi.org/10.12775/QS.2025.42.61192>

21. Muñoz-Cano, R., et al. (2018). Impact of allergic rhinitis on quality of life, sleep and anxiety: A Spanish study. *Clinical and Translational Allergy*, 8, 23. <https://doi.org/10.1186/s13601-018-0212-0>
22. Papapostolou, G., Kiotseridis, H., Romberg, K., Dahl, Å., Bjermer, L., Lindgren, M., Aronsson, D., Tunsäter, A., & Tufvesson, E. (2021). Spatial working memory and executive function in children with allergic rhinitis. *Pediatric Allergy and Immunology*, 32, 67. <https://doi.org/10.1111/pai.13328>
23. Rollema, C., Metting, E. I., & Vries, T. W. (2022). Pharmacological treatment of allergic rhinitis: A narrative review. *Clinical & Experimental Allergy*, 52(10), 1247–1262. <https://doi.org/10.1111/cea.14114>
24. Sirohi, P., Khan, M. H., Sharma, S., Nuhmani, S., Al Muslem, W. H., & Abualait, T. (2022). A systematic review of effects of daytime napping strategies on sports performance in physically active individuals with and without partial-sleep deprivation. *PeerJ*, 10, e14460. <https://doi.org/10.7717/peerj.14460>
25. Song, Y., Huang, Y., Gao, Y., Zhang, M., Shao, Y., Zhou, G., Sun, H., Wang, G., Jia, T., Shi, J., & Sun, Y. (2025). Stress management in athletes: Predictive effects of sleep deprivation-induced cognitive control changes on competition performance. *The Journal of Neuroscience*, 45(16), e1683242025. <https://doi.org/10.1523/JNEUROSCI.1683-24.2025>
26. Sun, X., Qu, Z., Zhang, X., Zhang, Y., Zhang, X., Zhao, H., & Zhang, H. (2025). The effects of sleep deprivation on cognitive flexibility: A scoping review of outcomes and biological mechanisms. *Frontiers in Neuroscience*, 19, 1626309. <https://doi.org/10.3389/fnins.2025.1626309>
27. Tabata, K., Sumi, Y., Sasaki, H., & Kojimahara, N. (2025). Effectiveness of intranasal corticosteroids for sleep disturbances in patients with allergic rhinitis: A systematic review and meta-analysis. *International Archives of Allergy and Immunology*, 186, 330–344. <https://doi.org/10.1159/000541389>
28. Vitale, K. C., Owens, R., Hopkins, S. R., & Malhotra, A. (2019). Sleep hygiene for optimizing recovery in athletes: Review and recommendations. *International Journal of Sports Medicine*, 40(8), 535–543. <https://doi.org/10.1055/a-0905-3103>
29. Wannell, B. R., Brunner, F. M. J., Lovibond, Z., & Bond, B. O. (2025). The influence of partial sleep restriction on repeated sprint ability and reaction time in university athletes. *Frontiers in Sports and Active Living*, 7, 1519987. <https://doi.org/10.3389/fspor.2025.1519987>
30. Yan, X., Liu, P., Wang, W., Xuan, J., Shen, M., & Peng, J. (2025). Causal effects of different types of physical activity on allergic rhinitis: A Mendelian randomization study. *Medicine*, 104(35), e44159. <https://doi.org/10.1097/MD.00000000000044159>
31. Zaccarin, M., Zanni, S., Gallè, F., Protano, C., Valeriani, F., Liguori, G., Romano Spica, V., & Vitali, M. (2022). Studying respiratory symptoms related to swimming pools attendance in young athletes: The SPHeRA study. *Toxics*, 10, 759. <https://doi.org/10.3390/toxics10120759>